

Organic and Organometallic Chemistry at the Single-Molecule, -Particle, and -Molecular-Catalyst-Turnover Level by Fluorescence Microscopy

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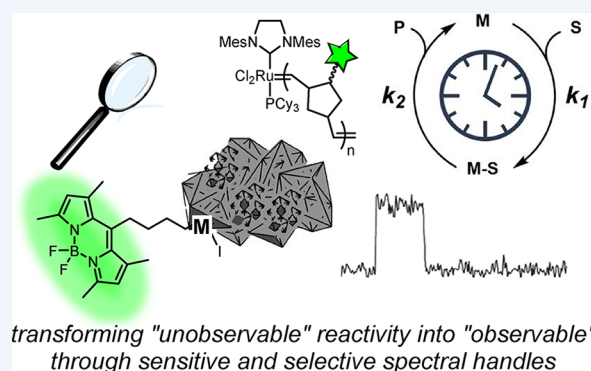
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CONSPECTUS: Mechanistic studies have historically played a key role in the discovery and optimization of reactions in organic and organometallic chemistry. However, even apparently simple organic and organometallic transformations may have surprisingly complicated multistep mechanisms, increasing the difficulty of extracting this mechanistic information. The resulting reaction intermediates often constitute a small fraction of the total reaction mixture, for example, creating a long-term analytical challenge of detection. This challenge is particularly pronounced in cases where the positions of intermediates on the reaction energy surface mean that they do not “build up” to the quantities needed for observation by traditional ensemble analytical tools. Thus, their existence and single-step elementary reactivity cannot be studied directly. New approaches for obtaining this otherwise-missing mechanistic information are therefore needed.

Single-turnover, single-molecule, single-particle, and other subensemble fluorescence microscopy techniques are ideally suited for this role because of their sensitivity and spatiotemporal resolution. Inspired by the robust development of single-molecule fluorescence microscopy tools for studying enzyme catalysis, our laboratory has developed analogous fluorescence microscopy techniques to overcome mechanistic challenges in synthetic chemistry, with sensitivity as high as the single-complex, single-turnover, and single-molecule level. These techniques free the experimenter from the previous restriction that intermediates must “build up” to quantities needed for detection by ensemble analytical tools and are suited to systems where synchronization through flash photolysis or stopped flow would be inconvenient or inaccessible. In this process, the techniques transform certain previously “unobservable” intermediates and their elementary single-step reactivities into “observable” ones through sensitive and selective spectral handles. Our program has focused on imaging reactions in small-molecule, organic, and polymer synthetic chemistry with an accent on the reactivity of molecular transition metal complexes and catalysts.

Our laboratory initiated studies in this area in 2008 with the imaging of individual palladium complexes that were tagged with spectator fluorophores. To enable imaging, we started with fluorophore selection and development, overcame challenges with imaging in organic solvents, and developed strategies compatible with air-sensitive chemistry and concentrations of reagents generally used in small-molecule synthesis. These studies grew to include characterization of previously unknown organometallic intermediates in the synthesis of organozinc reagents and the direct study of their elementary-step reactivity. The ability to directly observe this behavior generated predictive power for selecting salts that accelerated organozinc reagent formation in synthesis, including salts that had not yet been reported synthetically. In 2017 we also developed the first single-turnover imaging of molecular (chemo)catalysts, which through the technique’s spatiotemporal resolution revealed abruptly time-variable polymerization kinetics wherein molecular ruthenium ring-opening metathesis polymerization (ROMP) catalysts changed rates independently from other catalysts less than 1 μm away. Individual catalytic turnovers, each corresponding to one single-chain-elongation reaction arising from insertion of single ROMP or enyne monomers at individual Grubbs II molecular ruthenium catalysts, were spatiotemporally resolved as green flashes in growing polymers.

In this Account, we discuss the development of this technique from idea to application, including challenges overcome and strategies created to image synthetic organic and organometallic molecular chemistry at the highest levels of detection sensitivity. We also describe challenges not yet solved and provide an outlook for this growing field at the intersection of microscopy and synthetic/molecular chemistry.



INTRODUCTION

Mechanistic studies of reactions in synthetic chemistry have historically been limited by the unsynchronized nature of thermal chemical reactions. This lack of synchronization prevents the observation of certain reaction intermediates,

making them and their elementary-step reactivities “unobservable” by direct methods. Specifically, an intermediate’s location

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on the reaction free energy surface controls whether it “builds up” to sufficient quantities for detection by traditional ensemble analytical techniques. This restriction is particularly limiting in mechanistic studies of transition metal reactions, including catalysis: given the range of potential reaction intermediates and the prevalence of multistep transformations, synthetically useful transformations proceeding through such “unobservable” intermediates and elementary-step reactivities are painfully (to the mechanistic chemist) common.

In 2005, one of us (S.A.B.) had completed a Ph.D. in mechanistic organometallic chemistry in a chemistry department¹ and was a postdoctoral scholar in mechanistic enzymology at a medical school. This cross-disciplinary postdoctoral training resulted in an opportunity to attend a seminar by Vale detailing the then-recent fluorescence imaging of single motor proteins walking hand-over-hand along microtubules.² This seminar provided a case study of the current status of single-molecule fluorescence microscopy, enough for a classically trained chemist to realize that many physical-organic technique limitations of probing chemical reaction mechanisms were analytical limitations similar to those being solved by single-molecule biophysicists.^{3,4} Thus, it should be possible to remove the similar ensemble averaging that complicated mechanistic investigations in chemistry. However, in 2005 no examples of single-molecule fluorescence imaging of chemical reactions existed in the broad area of what might be considered synthetic chemistry or catalysis. Just outside this area there were a handful of experiments characterizing the chemical behavior of dyes themselves at the single-molecule level, e.g., photoisomerization⁵ and photobleaching of an aromatic hydrocarbon⁶ and of a phosphorescent organoiridium complex.⁷ Presumably, the absence of prior reports combining single-molecule fluorescence microscopy with questions in synthetic chemistry and catalysis arose from a lack of discussion between synthetic chemists, who knew what the interesting mechanistic problems were, and biophysicists or physical/analytical chemists, who knew the current abilities of the technique. Here was an opportunity to develop a technique to solve some of the mechanistic questions in synthetic chemistry that had previously been just out of reach. To have this realization while sitting in Vale’s seminar was very exciting.

In this Account, we describe our research group’s development of fluorescence microscopy as a tool for investigations in synthetic chemistry and molecular catalysis.^{8–24} Time showed that our laboratory was not the only one to take an interest in developing fluorescence microscopy techniques to obtain information on chemical systems, and we eventually benefited from inspiration and reports from other researchers. These other researchers initially focused on the reactivity of nanoparticles and zeolites or modified DNA–metal complexation, areas that were orthogonal to our research in synthetic chemistry but had a central shared interest in chemical reactivity.^{25–28} A timeline of contributions by others that we found particularly interesting, relevant, or thought-provoking is provided at the end of the Account to supply the reader with a wider perspective of the field’s development.

Definitions here set the scope of this Account. The term “subensemble” refers to a collection of molecules that is sufficiently small to permit observation of unique (that is, non-ensemble-averaged) chemical behavior. Single-molecule and single-turnover studies described here are examples of subensemble studies at the ultimate sensitivity limit of single small molecules and molecular catalysts. Our studies have focused on small-molecule organic/organometallic chemistry

and related molecular metal catalysis (i.e., topics in “synthetic organic and organometallic chemistry”). Given the water solubility of biomolecules and their strong, selective binding affinities, we quickly found that experimental challenges that occur when investigating their behavior are distinct from those that occur when investigating synthetic chemistry. For example, a technique for investigating small-molecule/molecular-catalyst synthetic chemistry would have to proceed with organic solvents, be able to visualize chemical change without the benefit of enzyme-like binding affinities and substrate/catalyst specificity, and be able to accommodate requirements for oxygen- and moisture-free conditions consistent with the realities of these reaction processes. These were some of the challenges that we faced during our initial cross-disciplinary technique development.

■ THE START OF METHOD DEVELOPMENT

The official start of our research laboratory at the University of California, Irvine, in 2006 provided a venue for considering these challenges and developing suitable techniques for imaging chemical reactions in the laboratory. We hypothesized that a spectator fluorophore approach, wherein transition metal complexes, substrates, or functional groups of interest could be tagged with an inert and remote fluorophore to enable imaging, would provide a high level of generality: a fluorophore tag could be attached remotely to plausibly almost any ligand–metal complex or small-molecule substrate. We refer to probes of this strategy as “spectators”, in contrast to “participants”, which react at the fluorophore itself.²⁹ Such a hypothesis hinged on identifying, synthesizing, and developing fluorophores that met four basic criteria: (1) they were not reactive, including no acidic protons or Lewis basic lone pairs that would bind or otherwise interfere with transition metal chemistry (properties particularly hard to find in commercial fluorophore conjugates, which are often designed for solubility in water, e.g., for labeling of biomolecules); (2) they were soluble in organic solvents; (3) they had high quantum yields; and (4) they could be synthesized on at least the tens of milligrams scale, and preferably higher, while tolerating a wide range of reaction conditions and remote functional groups. This scale may seem small to scientists outside this area, but it is quite large relative to the quantities of some commercially available designer fluorophores for biological microscopy applications (some in microgram quantities).

■ PRACTICAL CONSIDERATIONS: DEVELOPMENT OF THE SPECTATOR FLUOROPHORE APPROACH, FROM GENERATING IMAGING AGENTS TO IMAGING SINGLE MOLECULAR TRANSITION METAL COMPLEXES

A misperception that our group encountered at the proposal stage was that a fluorescence microscopy approach would be fundamentally incompatible with studying transition metal chemistry. This misperception arose from the belief that transition metal complexes would necessarily quench all fluorescence, rendering all chemistry unimageable. Literature precedent, however, had already established that at least some transition metal complexes with fluorescent ligands retained fluorescence.^{30,31} Nevertheless, the misperception of incompatibility would persist until we could demonstrate the first single-molecule imaging of a transition metal complex tagged with a fluorophore that retained its fluorescence—a central premise of

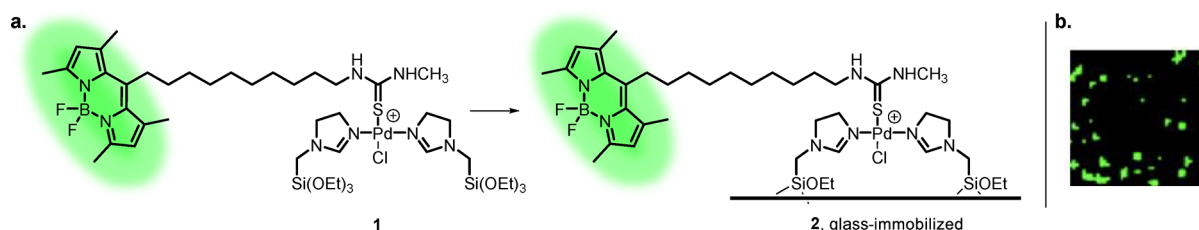


Figure 1. Imaging of single palladium complexes with spectator fluorophores. (a) Strategy of immobilizing molecular complexes on glass. (b) Imaging data for 2 after reaction. Single palladium complexes appear as green spots. The image in (b) was reproduced from ref 8. Copyright 2008 American Chemical Society.

the method. We thus set out to establish the plausibility of this central premise and to test the generality of the approach.

We selected boron dipyrromethene (BODIPY) dyes³² as our target fluorophore class, anticipating that we could develop them to meet the needs of the research program. We briefly tried perylene diimide fluorophores but found their requisite synthetic intermediates to be prohibitively insoluble for our studies. In our first development of BODIPY fluorophores as general spectator imaging agents for transition metal complexes, our nascent group synthesized palladium complex 1.⁸ This complex contained a green BODIPY fluorophore separated from the palladium center by an inert 10-carbon alkyl spacer (Figure 1a). Looking at 1 now, the first aspect we notice is the long spacer, a homage to early thoughts about the metal center quenching the fluorescence if the units were too close. The long spacer produced messy alkyl regions in NMR spectra and floppy, greasy molecules with difficult purification. We later discovered that the BODIPY unit remained sufficiently fluorescent for single-molecule imaging when closer to our metal centers of interest. Therefore, we shrank the chain for practical reasons in future studies. Immobilization of 1 via silyloxy chemistry on a glass microscope coverslip produced the first example of a spectator-fluorophore-tagged transition metal imaged at the single-molecule level. Individual complexes of 2 were imaged after the reaction as bright green spots on the otherwise dark glass surface (Figure 1b).⁸

IMAGING CHEMICAL REACTIONS AT SINGLE MOLECULAR TRANSITION METAL COMPLEXES

The next step in development was to image a chemical reaction as it occurred. Because our spectator fluorophores did not change chemically during reactions, their fluorescent emission color was constant by design. Thus, we needed something other than changing spectral properties to signal chemical reactivity. After consideration for potential generality and with inspiration from biophysical imaging examples, we decided to develop strategies to harness the change in *location* of the fluorophore-tagged molecule to obtain information on the occurrence of a chemical reaction remote to the fluorophore at a (possibly diverse) functional group/metal complex of interest.³³ Under our selected total internal reflection fluorescence (TIRF) microscopy conditions, molecules freely diffusing in solution appear only as a background haze because they move at a rate faster than the imaging rate.³⁴ Furthermore, in TIRF the laser excitation angle exclusively allows excitation of molecules near/on the surface of the glass coverslip. Thus, the bulk of molecules in solution remain dark, producing low backgrounds and high sensitivity for imaging of single molecules attached to the surface.³⁴

We developed this TIRF strategy (essentially invisible when in solution but bright when on the surface) to image the first

reactions at individual molecular transition metal complexes (Figure 2).^{16,17,19} Each ligand exchange reaction brought a

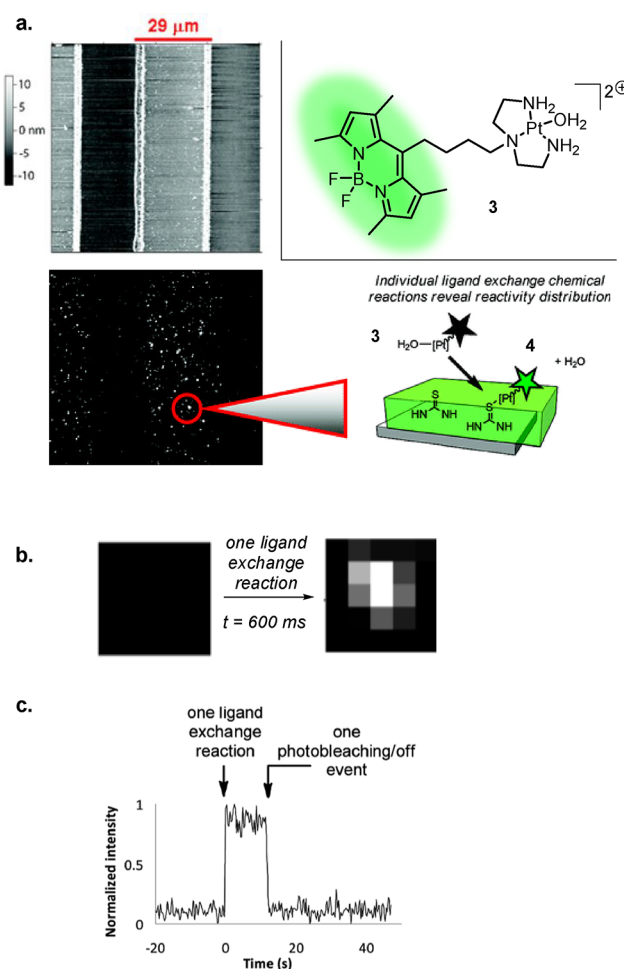


Figure 2. Individual ligand exchange reactions. (a) Photopatterned striped surface in AFM and fluorescence. Each white dot is one spatiotemporally resolved reaction. (b) Zoomed-in view of one ligand exchange reaction (267 nm pixel size). (c) Quantized fluorescence signal vs time from one ligand exchange reaction. Reproduced from (a) ref 16 and (b, c) ref 19. Copyright 2010 and 2011, respectively, American Chemical Society.

single complex of 3 from solution onto the thiourea surface to form complex 4, producing a spatiotemporally resolved bright green flash on the surface upon reaction. Each such event produced a quantized/stepwise increase in fluorescence intensity, observable as a jump in the fluorescence versus time trace at a single location in the sample, followed by a quantized decrease in fluorescence intensity upon photobleaching, which

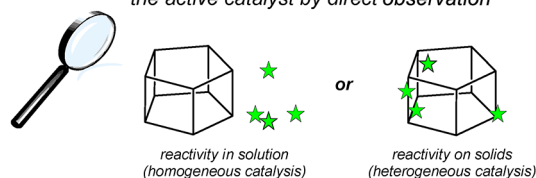
are well-established fingerprints of single molecules³⁵ (Figure 2c). Over the course of about 10 s, tens of individual ligand exchange reactions occurred; each was marked at the time it occurred¹⁷ by the appearance of a unique fluorescence signal.

Photolithography enabled comparison of stripes of thiourea-modified glass adjacent to unmodified glass in the same experimental sample and established that the reactivity was selectively localized on the thiourea surface. Individual chemical reactions could be localized below the diffraction limit (i.e., "superlocalized") to ± 11 nm.¹⁹ The ability to image ligand exchange, a broadly shared transformation in stoichiometric and catalytic metal chemistry,³⁶ supported the viability of our approach.

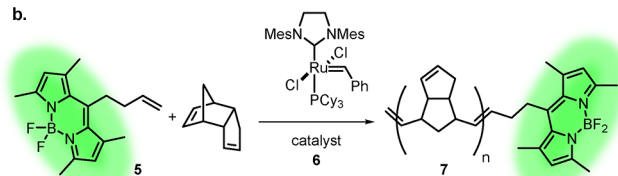
OVERCOMING A MECHANISTIC/ANALYTICAL CHALLENGE IN SYNTHESIS: DETERMINATION OF THE PHASE OF AN ACTIVE CATALYST BY DIRECT OBSERVATION

Improving catalytic processes often hinges on understanding the nature of the active catalyst.³⁶ Traditional ensemble analytical techniques are well-suited to determining the presence of major components in a reaction mixture, but the simple presence of solids or soluble components does not imply their catalytic activity, leading to a gap in analytical ability. We considered that it might be possible to develop a microscopy method to determine the phase of the active catalyst by direct observation, i.e., by looking at the solids or in the solution and directly imaging the reaction location (Figure 3a).

a. A molecular "who done it" mystery: determining the phase of the active catalyst by direct observation



b.



c.

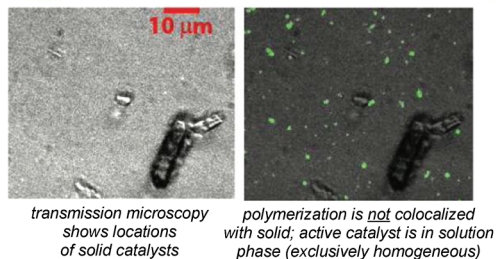


Figure 3. Determination of the phase of an active catalyst by direct observation: (a) schematic; (b) reaction; (c) microscopy data. The images in (c) were reproduced from ref 18. Copyright 2011 American Chemical Society.

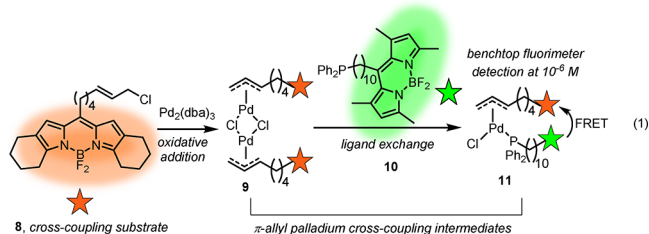
In 2011 we reported imaging agent 5, which informed on chain termination events during ring-opening metathesis polymerization (ROMP) reactions of dicyclopentadiene catalyzed by molecular ruthenium complex 6 (Grubbs II) (Figure 3b).¹⁸

The location of catalysis was determined by the location of green fluorescent polymers 7. Solid (undissolved) crystals of 6 were identifiable under these conditions, enabling monitoring of their surfaces for polymerization activity during the catalytic reaction. The ability to detect single-molecule reactions provided a high level of sensitivity for observing potential surface reactivity, yet no catalytic activity was detected on the surfaces of solid 6 (Figure 3c). This observation led to the determination that the reaction was exclusively catalyzed by solution species (homogeneous catalysis) without observable contribution by the solids (heterogeneous catalysis). In 2014, Scaiano applied a similar approach to determine the phase of an active catalyst by observing that single click reactions occurred on the surface of copper nanoparticles, showing that this imaging approach could be generalized to different chemical systems.^{37,38}

We also reported crossover and pulse–chase experiments via two-color green and orange BODIPY-tagged olefins to determine the mechanistic origin of the polymer morphology and the relative rates of different steps.²¹ Additional classical physical-organic mechanistic experiments likely have forthcoming single-molecule/single-particle congeners.

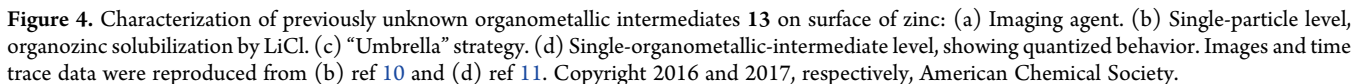
THWARTED BY SYNTHESIS: A FRET APPROACH TO STUDY π -ALLYLPALLADIUM CROSS-COUPLING REACTIONS THAT NEVER MADE IT TO THE SINGLE-MOLECULE LEVEL

At this time, we were interested in studying elementary steps during palladium catalysis using a two-color/two-ligand spectator fluorophore approach.^{9,23} As steps toward these studies, we synthesized orange allyl chloride oxidative addition substrate 8 en route to catalytically relevant³⁹ orange fluorescent π -allylpalladium complex 9 and orange-and-green 11 (eq 1).



We established that the formation of 11 upon titration of 9 with phosphine ligand 10 could be quantitatively characterized on the ensemble level with a benchtop fluorimeter at concentrations as low as 10^{-6} M. Analysis of a similar sample by ³¹P NMR spectroscopy at the same low-micromolar concentration as the fluorescence measurement showed no detectable signals, highlighting the higher sensitivity of the fluorescence technique to characterize low loadings of catalytic intermediates that might be present in complicated mixtures during synthetic cross-coupling reactions.

We intended to study this reaction and other reactivities of complexes 9 and 11 at the single-molecule level. However, the multistep synthesis of these two-color fluorophore-tagged complexes was overly involved. After we consumed the original batch of material, produced by a postdoctoral scholar who had by then completed training, the level of expertise and patience needed for its synthesis meant that we moved on to other ideas. By including a frank discussion of these synthetic challenges, we hope to inspire new solutions in the community.



Our spectator strategy allows imaging of the reactivity of remote functional groups common in synthesis rather than being limited to the reactions of the fluorophores themselves. Finally, we have developed strategies to study commercial catalysts without chemical modification or glass immobilization and to study commercial metal powders with the ranges of shapes, sizes, and surface chemical compositions used in reported synthetic

reactions. These features add to the synthetic relevance of the data.

The experimentalist can therefore select an approach that offers the best match with the experiment at hand. Our techniques and assessment of relevance are further described in the case studies below.

■ PREVIOUSLY UNOBSERVED ORGANOMETALLIC INTERMEDIATES: MECHANISTIC DETERMINATION OF THE ROLE OF LITHIUM CHLORIDE IN PROMOTING ORGANOZINC REAGENT SYNTHESIS

The generation of organometallic reagents and catalysts via oxidative addition of organohalides to metal powders would be the most direct and atom-economical synthesis of a wide range of organometallics, yet this path currently works well for few metals (e.g., organomagnesium or “Grignard” reagents). In 2006, Knochel reported that the simple salt lithium chloride promotes the formation of organozinc reagents directly from organohalides and zinc metal powder.⁴⁶ The mechanistic role of the salt was unknown, however. This lack of knowledge prevented a targeted approach to finding additives that promote direct insertion reactions to other metals broadly. We became interested in this mechanistic question and wondered whether microscopy approaches could be developed in order to answer it.

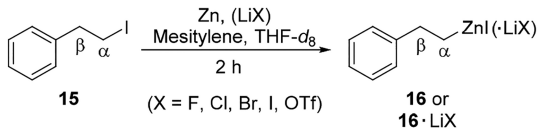
An advantage of fluorescence microscopy to solve this challenge would be that, in theory, imaging could occur in situ with the irregular zinc metal powder of the same supplier and mesh from the original synthetic reaction.⁴⁶ That tantalizing possibility contrasted with 1980s experiments by Whitesides to study Grignard reagent formation, which required single-crystalline magnesium that was spark-cut and chemically polished to reveal smooth surfaces and analyzed ex situ by scanning electron microscopy after reaction completion.⁴⁷ However, our system would require imaging chemical reactions on the surfaces of irregular, optically opaque, and large metal particles—too large to use TIRF to block out the fluorescence background. Epifluorescence mode, where the excitation laser points straight into the solution, creating higher background, would have to be employed. We considered that whether a fluorescence approach could work in this “messy” system would be a decent proving ground for applications in synthesis more generally.

In-house-designed imaging agent **12** containing a bright-green chemically inert BODIPY fluorophore, butyl spacer, and insertion-reactive carbon–iodide bond, was added to commercial zinc powder in THF in the absence of additives (at 2 mM), with rinsing of excess substrate after 20 h (Figure 4a,b).¹⁰ Previously unknown persistent surface organozinc intermediate **13** was detected on the zinc surface as bright green “hot spots” (Figure 4b). Each particle, and different surface regions of the same particle, were distinctive in their chemical and physical properties, as observed by the heterogeneous dispersion of **13**. The “hot spots” in Figure 4b contained many molecules of imaging agent, but we also detected single molecules of these intermediates under reduced concentration conditions (Figure 4d).¹¹ The “umbrella” shadow that shielded **13** from the background haze produced a sufficiently dark background for single-molecule imaging (Figure 4c), despite the high background in epifluorescence mode (green haze in Figure 4b,d). This was the first report of imaging single organometallic intermediates via fluorescence microscopy.

Addition of LiCl resulted in removal of the “hot spots” of **13** to form solution organometallic reagent **14**. Thus, the mechanistic

role of LiCl was determined to be solubilization of otherwise persistent surface intermediate **13**. Parallel ¹H NMR and ⁷Li NMR spectroscopy studies showed that the solubilization behavior of intermediate **13**, observed by microscopy with different lithium salts, was correlated with reaction acceleration in the bulk synthesis of **16**. This correlation enabled *synthetic prediction* of which lithium salts would accelerate bulk reagent synthesis of **16** from **15**, including those that had not yet been reported, solely on the basis of the salt’s effect on solubilization of intermediates **13** observed by microscopy (Table 1).²⁴

Table 1. Salt Effect on Intermediates Observed by Fluorescence Microscopy Predicts Macroscale Synthesis Effects²⁴

		
salt	NMR conversion to 16 [%]	salt solubilizes surface intermediate 13 by microscopy
no salt	57	—
LiOTf	62	no
LiF	56	no
LiCl	99	yes
LiBr	99	yes
LiI	>99	yes

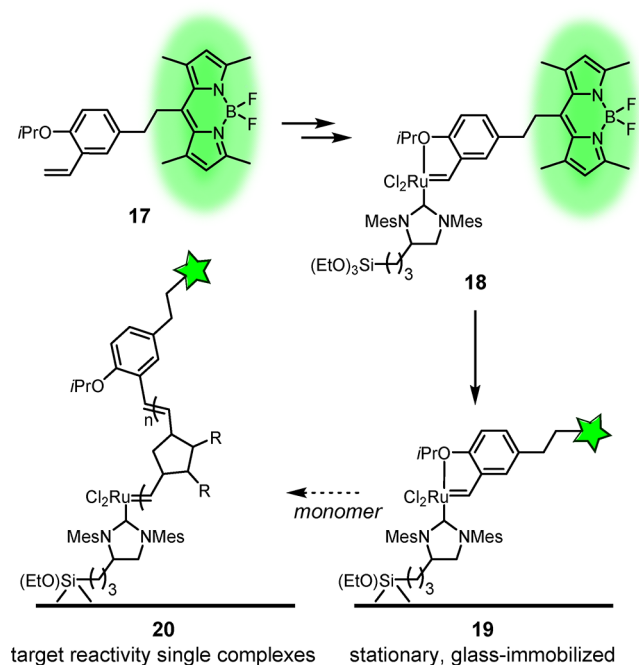


Figure 5. Initially targeted glass immobilization of single molecular polymerization catalysts.

■ SINGLE TURNOVER AT MOLECULAR CATALYSTS: RING-OPENING METATHESIS POLYMERIZATION WITH SINGLE-MONOMER/SINGLE-CHAIN-ELONGATION RESOLUTION

An often-cited advantage of molecular catalysts over solid extended-structure catalysts is their higher potential selectivity, arising from the uniformity of the coordination environment

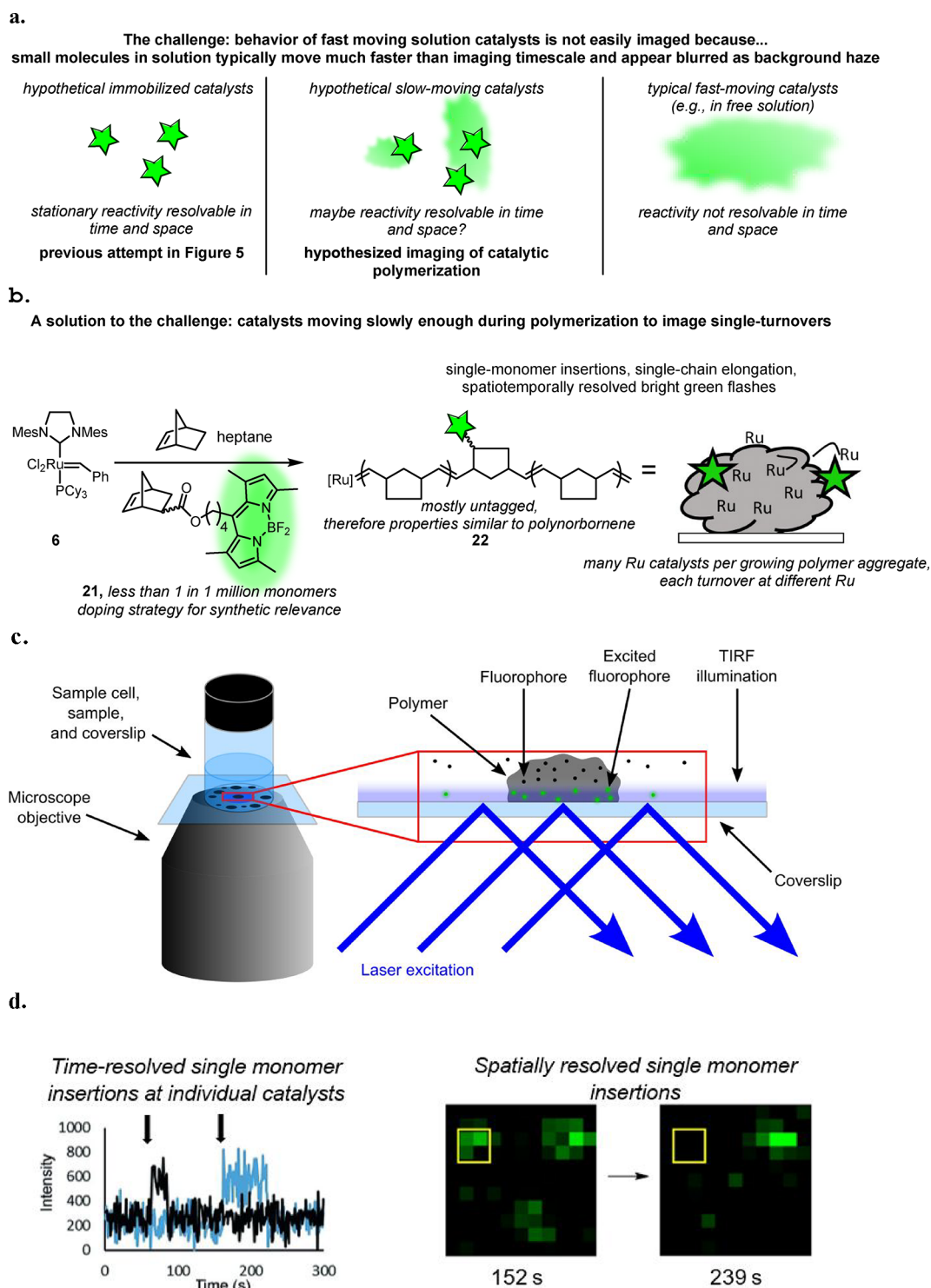


Figure 6. Single-turnover imaging of molecular polymerization catalysts. (a) Schematic of the challenge. (b) Schematic of the precipitation polymerization approach. (c) Schematic of microscopy imaging. (d) Data (each pixel 267 nm). The time trace data and microscopy images in (d) were reproduced with permission from refs 12 and 13. Copyright 2017 and 2018, respectively, Wiley-VCH.

around each catalyst.³⁶ But how similar are individual molecular catalysts to each other in reality? Single-molecule or single-particle studies in biology and with zeolites and nanoparticles have established that there are almost always variations between individuals within the ensemble. Our own studies showed reactivity distributions on zinc particles and solid supports. If these previous studies are our guides, then there are almost certainly molecular catalyst systems with high individual

reactivity variation as well. The questions intriguingly become *when, where, how, and on what time scale*—rather than simply *if*.

We became interested in investigating whether single-molecule/single-turnover studies of molecular transition metal catalysts could reveal the characteristics of individuals. The challenge with imaging single molecular catalysts is that they typically diffuse at a faster rate in solution than the imaging rate. Finding a way to slow them down seemed like a plausible initial

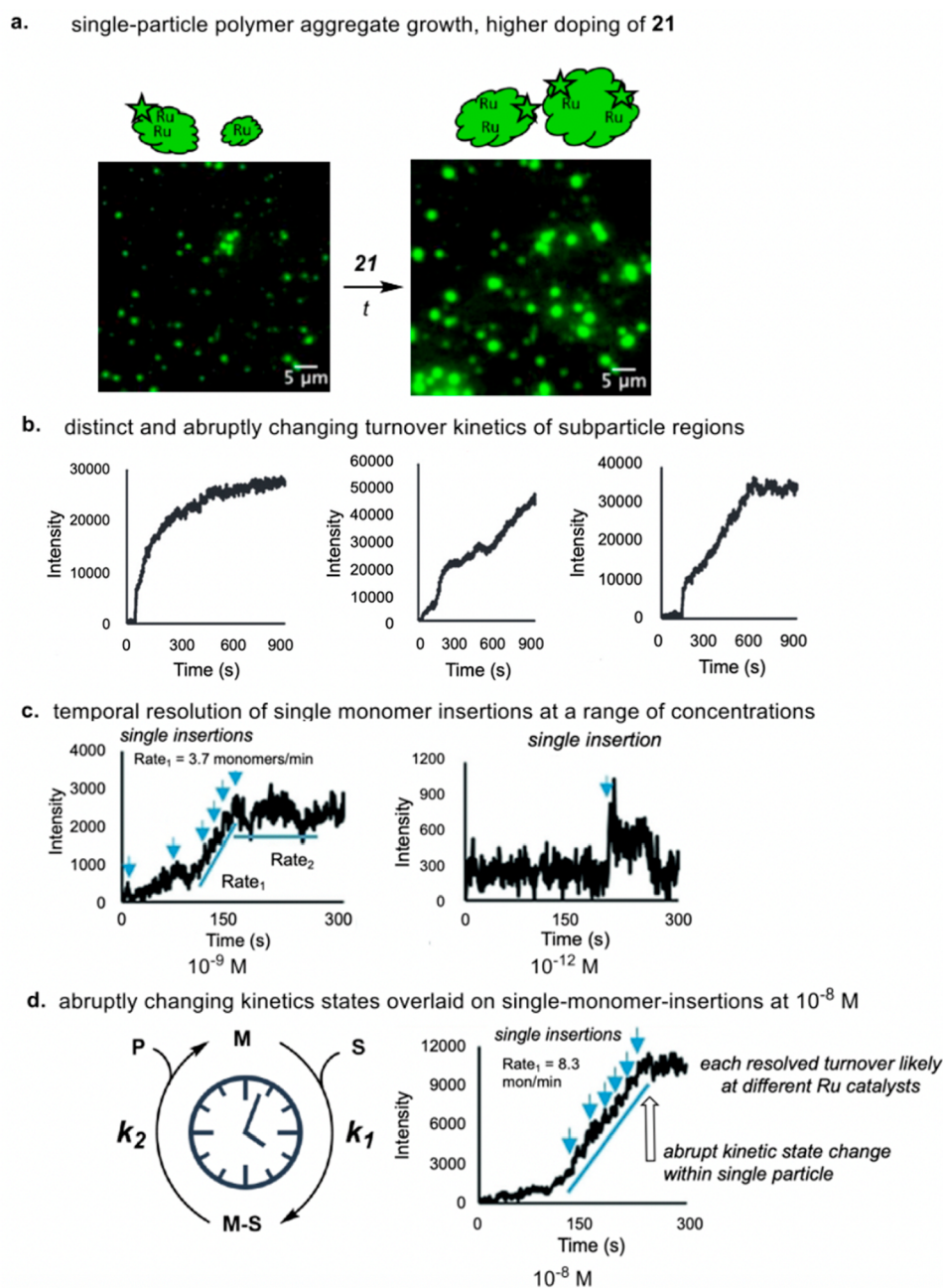


Figure 7. Imaging of unique (nonaveraged) kinetics. (a) Single particle aggregates grow larger and brighter. (b) Distinct and abruptly changing kinetics in different subparticle regions. (c) Temporal resolution of single catalytic turnovers at molecular catalysts as “steps” at a range of imaging agent concentrations. (d) Distinct kinetic states derived from overlay of single insertions into polynorbornene. The microscopy images in (a) and the time trace data in (b)–(d) reproduced with permission from refs 13–15. Copyright 2017 and 2018, respectively, Wiley-VCH.

approach for this imaging challenge. We first tried glass immobilization of BODIPY-tagged molecular ruthenium complex **18**, derived from in-house-designed and -synthesized imaging agent **17**, to study ROMP at the individual-catalyst level (Figure 5). After developing the requisite multistep synthesis and getting excited about imaging single molecular complexes of **19**, and an early polymerization hit plausibly resulting in **20**, we found the reactivity of this system to be finicky. We never published those microscopy studies, opting only to report the synthesis of the fluorescent synthetic precursor **17**.²³

Despite this initial challenge with a glass-surface-supported catalyst, we continued to be interested in polymerization catalysts. The changing local microenvironment caused by the

growing and potentially conformationally dynamic polymer⁴⁸ presented a plausible platform for generating distinct individual catalyst behaviors, though it was not clear under what conditions (if any) such individual behaviors would occur and could be detected. These individual molecular catalyst variations could, in theory, profoundly impact the synthetic outcome of the reaction, i.e., by underpinning (the lack of) catalytic reaction selectivity as observed on the bulk scale.

We considered the possibility that molecular catalysts need not be strictly stationary on the time scale of imaging but rather only need be moving quite slowly. If that were the case, then individual molecular catalysts attached to strands of growing polymers might be moving sufficiently slowly for imaging of

<i>From Other Laboratories</i>		<i>From Our Laboratory</i>
Single turnover at inorganic solids; ²⁵ participant (<i>Hofkens, Roelfaers</i>)	2006	
	2007	
Single-turnover at Au nanoparticles; ²⁶ participant (<i>Chen</i>)	2008	Single transition-metal complex imaging; ⁸ spectator
Photocatalysis at titanasilicate zeolite surface; ²⁷ participant (<i>Majima</i>)	2009	FRET ligand exchange Pd molecular complexes, ensemble; ⁹ spectator
	2010	Single ligand exchange molecular complexes; ¹⁶ spectator
Single-turnover imaging zeolite; ⁴¹ participant (<i>Roelfaers</i>)	2011	Determination of phase of catalysis, molecular Ru polymerization; ¹⁸ spectator Single ligand exchange superlocalization Pt molecular complexes; ^{17,19} spectator
	2012	
mCPBA epoxidation reaction pathways, molecular; ⁵³ participant (<i>Trapp, Jung, Herten</i>)	2013	Mechanistic origin of polymer morphology; ²¹ spectator
Fluorescence and electron microscopy nanoparticle correlation; ⁵⁴ participant (<i>Chen</i>)	2014	
Copper click phase of nanoparticle catalysis single molecule FRET; ³⁷ spectator (<i>Scaiano</i>)	2015	
Ruthenium metathesis products; ⁵⁵ participant (<i>Jung</i>)	2016	Organometallic intermediates and mechanism of organozinc/LiCl; ¹⁰ spectator
3D microscopy in porous zeolites; ⁵⁶ participant (<i>Roelfaers</i>)	2017	Single turnover at molecular catalysts; ¹² Single organometallic intermediates; ¹¹ spectator
Ligand initiation kinetics of molecular Pd catalysts; ⁴⁴ spectator (<i>Goldsmith</i>)	2018	Time-variable, non-averaged kinetics at molecular polymerization catalysts; ^{13,14} spectator
Single-molecule fluorescence and electron microscopy same instrument; ⁵⁷ participant (<i>Roelfaers</i>)	2019	Catalyst speciation, single-turnover enyne metathesis at molecular catalysts; ¹⁵ Microscopy prediction of bulk-scale synthetic effects; ²⁴ spectator

Figure 8. Timeline of selected advances.

their single-turnover reactivity naturally, without the need for chemical modification of the catalyst or other artificial immobilization strategies to create a stationary catalyst (Figure 6a). For synthetic relevance, we opted to study a precipitation polymerization process involving norbornene and commercial catalyst **6** by doping imaging agent **21** (at less than one in 1 million monomers) into up to 30 mM untagged norbornene monomer. The insertion of a single monomer of **21** by molecular catalyst **6** as the catalyst turned over during one unit of chain elongation/polymerization produced individual bright flashes of green light within the resulting growing single-particle polymer aggregates **22** (Figure 6b). The experiments observed the insertion of a single monomer by a single catalyst as the catalyst turned over during polymerization. Multiple single-turnover events were spatiotemporally resolved within single aggregates, each likely arising from a different molecular ruthenium catalyst (Figure 6c,d). Thus, individual ruthenium catalysts were sufficiently stationary in these growing polymers for spatiotemporal resolution of their single-monomer-insertion reactions. This outcome meant that single turnovers could be imaged with the unmodified commercial catalyst without the compounding kinetic effects of a glass surface, increasing the synthetic relevance of the data and the accessibility of the precursors needed for study.

At higher concentrations of imaging agent, incorporation of **21** was faster than photobleaching, leading to a buildup of the fluorescence signal as each polymer aggregate **22** grew (Figure 7a).

The increase in fluorescence signal provided an estimate of the kinetics of subparticle regions of growing polymers. Each region contained multiple ruthenium centers, but surprisingly not so many that the kinetic behavior was ensemble-averaged: individual regions showed distinct and abruptly changing turnover kinetics, even displaying apparently different kinetic orders within the same sample (e.g., pseudo-zeroth order and first order; Figure 7b). The mechanistic origin of this distribution of behavior is not yet fully understood, but it may arise from changes in the local microenvironments around individual catalysts caused by the growing polymers.^{15,48} Although changes in the rates of insertion at single catalysts could not be isolated from these data, the kinetic profiles nevertheless provided the first evidence that individual molecular catalysts exhibit changes in their turnover kinetics independently of each other and distinct from the averaged behavior in the bulk sample.

Single catalytic turnovers in growing polymers were further resolved as stepwise increases in fluorescence intensity in time traces from 10^{-8} to 10^{-12} M in imaging agent **21** and in a suite of related imaging agents (Figure 7c,d), including those for single enyne metathesis reactions.^{14,15} Figure 7c,d shows the overlay of single catalytic turnovers within polymer aggregates, each of which likely occurred at a different ruthenium center given the low doping of fluororophore-tagged monomer. The continued measurement of unique (i.e., non-ensemble-averaged) kinetics at 10^{-8} M is detectable through the sum of colocalized signals

from a small number of resolved single monomer insertion/single molecular catalyst turnover events at neighboring ruthenium centers that show a localized abrupt change in kinetic state (e.g., ~6–10 individual chemical reactions in Figure 7d). In Figure 7c, Rate₁ is equal to 3.7 monomers/min and abruptly decreases to the Rate₂ regime; in Figure 7d, Rate₁ is equal to 8.3 monomers/min before an abrupt analogous change to a slower regime.

Thus, imaging of single catalytic turnovers at multiple different individual molecular catalysts (e.g., spatiotemporally resolved turnovers at a single molecular catalyst [for one turnover] or different colocalized molecular catalysts [for 2–10 turnovers]; Figure 7, our reports) has been achieved, as has imaging of single catalyst initiation events at multiple individual different glass-immobilized molecular catalysts (Goldsmith), but not yet many resolvable individual turnovers at the same molecular catalyst.^{12–15,44,48} Though the future is notoriously hard to predict, we anticipate that this next level of resolution of multiple single turnovers at the same individual molecular catalyst will soon be reached.

TIMELINE

We here select advances that we found/find particularly relevant or interesting in the development of fluorescence microscopy to study chemical reactions (Figure 8). This list is not comprehensive. Of particular note is the seminal report in early 2006, before our laboratory started experiments, by Roelfaers and Hofkens on imaging catalytic turnover on the faces of inorganic solids in water using a participant/reactive profluorophore substrate.²⁵ The highlighted reports from Chen, Majima, and Roelfaers launched substantial research programs with many subsequent publications on nanoparticles, photocatalysts, and zeolites from these laboratories.^{40,42,49–52} We consider the single but impactful reports by Goldsmith⁴⁴ on molecular catalyst initiation and Trapp, Jung, and Herten⁵³ on mCPBA epoxidation mechanisms to be the examples most related to synthetic/mechanistic molecular chemistry from other groups. In the timeline, we note the approach of the researchers (i.e., participant/reactive or spectator/inert fluorophore). Together with Cordes, our laboratory previously described our thoughts on the appropriateness of the participant and spectator fluorophore approaches to different research problems, which may help the experimenter choose an approach suited to the questions at hand.²⁹

OUTLOOK

Mechanistic questions in synthetic chemistry that were previously out of reach because of ensemble averaging are accessible with this and related fluorescence microscopy techniques. A very real possibility exists to rewrite our dogma on what types of intermediates and chemical behaviors in synthetic organic and organometallic chemistry are “observable” and to determine the impact of the newly revealed reactivity on bulk-scale chemistry.

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Notes

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Suzanne A. Blum received a B.S. in Chemistry at the University of Michigan, working with Prof. Edwin Vedejs, and a Ph.D. from the University of California, Berkeley, under the supervision of Professors Robert G. Bergman and Jonathan A. Ellman. After an NIH Postdoctoral Fellowship with Professor Christopher T. Walsh at Harvard Medical School, she joined the Chemistry Department at the University of California, Irvine, in 2006. Her research interests include mechanistic and synthetic studies of chemical reactions.

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